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Advanced Control System Design - Tutorial 4 – Solution

Note that some steps have been skipped. The complete steps can be found in the Lecture's slides and Tutorial 1 Solution (for matrices inversion).

1. Find the transfer function of the following state-space system:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-9 & -26 & -24\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}u$$

$$y=\begin{bmatrix}0 & 0 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Solution:

$$G(s)=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+D$$

$$G(s)=\begin{bmatrix}0 & 0 & 2\end{bmatrix}\begin{bmatrix}s+9 & 26 & 24\\ -1 & s & 0\\ 0 & -1 & s\end{bmatrix}^{-1}\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}+0$$

$$G(s)=\begin{bmatrix}0 & 0 & 2\end{bmatrix}\left[\frac{1}{s^3+9s^2+26s+24}\begin{bmatrix}s^2 & -2(13s+12) & -24s\\ s & s(s+9) & -24\\ 1 & s+9 & s^2+9s+26\end{bmatrix}\right]\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$$

$$G(s)=\frac{2}{s^3+9s^2+26s+24}$$

2. Perform diagonal transformation on the following state-space equations:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -12 & -16 & -7\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 5\end{bmatrix}u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 12 & 16 & \lambda + 7 \end{vmatrix} = \lambda^3 + 7\lambda^2 + 16\lambda + 12 = (\lambda + 2)(\lambda + 2)(\lambda + 3)$$

Repeated real roots. Thus:

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ \lambda_1 & 1 & \lambda_3 \\ \lambda_1^2 & 2\lambda_1 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 4 & -4 & 9 \end{bmatrix}$$

$$\dot{\mathbf{z}} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{z} + \mathbf{S}^{-1}\mathbf{B}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -3 & -4 & -1 \\ 6 & 5 & 1 \\ 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -12 & -16 & -7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 4 & -4 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} -3 & -4 & -1 \\ 6 & 5 & 1 \\ 4 & 4 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} -5 \\ 5 \\ 5 \end{bmatrix} u$$

$$y = \mathbf{C}\mathbf{S}\mathbf{z} + \mathbf{D}u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 4 & -4 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + 0 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$